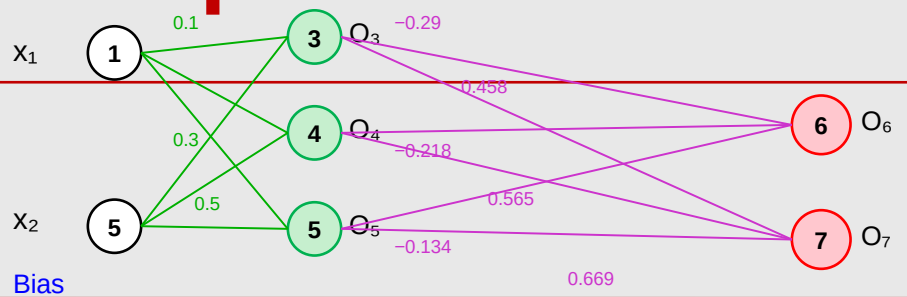
Example

$$\delta_j = O_j(1 - O_j)\left[\sum_k (\delta_k w_{kj})\right]$$

$$\Delta w_{ji} = \eta O_i O_j (1 - O_j)\left[\sum_k (\delta_k w_{kj})\right] = \eta O_i \delta_j$$



$\delta_6 = 0.791(1-0.791)(0-0.791) = -0.131$ $\delta_7 = 0.791(1-0.791)(1-0.791) = 0.035$

$\eta = 5$

Backward (Update hidden Layer Weights)

$\delta_3 = O_3(1-O_3)[\delta_6 * W_{63} + \delta_7 * W_{73}]$

$\delta_3 = 0.9(1-0.9)[-0.131*-0.29 + 0.035*0.458] = 0.005$

$\delta_4 = 0.943(1-0.943)[-0.131*-0.218 + 0.035*0.565] = 0.003$

$\delta_5 = 0.968(1-0.968)[-0.131*-0.134 + 0.035*0.669] = 0.001$

$\Delta W_{31} = 5*0.005*1 = 0.025$ $W_{31} = 0.1 + 0.025 = 0.125$

$\Delta W_{32} = 5*0.005*5 = 0.125$ $W_{32} = 0.4 + 0.125 = 0.525$

$\Delta W_{3b} = 5*0.005*1 = 0.025$ $W_{3b} = 0.1 + 0.025 = 0.125$

$\Delta W_{41} = 5*0.003*1 = 0.015$ $W_{41} = 0.2 + 0.015 = 0.215$

$\Delta W_{42} = 5*0.003*5 = 0.075$ $W_{42} = 0.5 + 0.075 = 0.575$

$\Delta W_{4b} = 5*0.003*1 = 0.015$ $W_{4b} = 0.1 + 0.015 = 0.115$

$\Delta W_{51} = 5*0.001*1 = 0.005$ $W_{51} = 0.3 + 0.005 = 0.305$

$\Delta W_{52} = 5*0.001*5 = 0.025$ $W_{52} = 0.6 + 0.025 = 0.625$

$\Delta W_{5b} = 5*0.001*1 = 0.005$ $W_{5b} = 0.1 + 0.005 = 0.105$

$$O_3 = \frac{1}{1+e^{-8.2}} = 0.9$$

$$O_4 = \frac{1}{1+e^{-2.8}} = 0.943$$

$$O_5 = \frac{1}{1+e^{-3.4}} = 0.968$$

$$O_6 = \frac{1}{1+e^{-1.3312}} = 0.791$$

$$O_7 = \frac{1}{1+e^{-1.3312}} = 0.791$$